Manish Nagaraj

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Objective: *A PhD candidate with extensive expertise in data efficiency and optimization techniques, focusing on creating compact, accurate, and scalable machine learning systems. Passionate about reducing computational overhead while enhancing system performance for real-world applications.*

Education

- **Purdue University** — West Lafayette, IN
  PhD in Electrical & Computer Engineering — *Expected Graduation: May 2026*
  - **Relevant Areas**: Data Efficiency, Efficient Fine-tuning, Federated Learning, Large Language Models
  - **Research Advisor**: Prof. Kaushik Roy
- **Purdue University** — West Lafayette, IN
  MS in Electrical & Computer Engineering — *May 2019*
  - **Thesis**: Energy Efficient Byzantine Agreement Protocols for Cyber-Physical Resilience
- **PES Institute of Technology and Science** — Bangalore, India
  Bachelor of Engineering in Electronics and Communications — *May 2017*

Research Projects

- **Exploring Data Efficiency in Deep Learning Systems** — West Lafayette, IN
  Ph.D. Dissertation, Purdue University — *May 2019 – Present*
  - **Data-Efficient Fine-tuning of Large Language Models** – *TRIM* (under review at ARR)
    - Proposed a **forward-only, gradient-free** data selection method that builds attention-based token "fingerprints" to score **instruction-tuning** examples.
    - Selected instruction examples that most improved performance on **reasoning and commonsense benchmarks** for **open-source LLMs** under small data budgets.
    - TRIM-selected **coresets** outperformed state-of-the-art coreset baselines by up to ∼9% on downstream tasks and, in some settings, matched or surpassed **full-data fine-tuning**.
    - Demonstrated **computational scalability**: in a shared coreset selection setup, TRIM achieved higher downstream accuracy while running ∼2.6× faster than the best gradient-based baseline.
  - **Identifying Coresets from Loss Trajectories** – *CLD* (TMLR 2025)
    - Introduced **Correlation of Loss Differences (CLD)**, a **gradient-free** coreset metric that ranks training points by how closely their **loss-change trajectories** correlate with a small validation set.
    - Established a **convergence guarantee**: training on CLD-selected coresets tracks full-data optimization up to a provable error bound.
    - On **CIFAR-100** and **ImageNet-1k**, **achieved state-of-the-art** coreset selection performance, typically within ∼1% of more computationally expensive methods across subset sizes.
    - Demonstrated **cross-architecture transfer**: coresets selected with small proxy CNNs generalized to larger **CNN** and **vision-transformer** models with $< 1\%$ accuracy drop.
  - **Privacy- and Communication-Efficient Federated Learning** – *TOFU* (IEEE Access 2024)
    - Proposed a **federated learning** scheme that encodes each client's model update as gradients on a small **synthetic dataset**, transmitting proxies instead of full weight updates.
    - On **MNIST** and **CIFAR-10**, achieved up to $\sim 4\times$ and $\sim 6.6\times$ lower **communication cost** than a standard federated learning baseline, while maintaining comparable final accuracy.
    - Enhanced **privacy** against **gradient inversion attacks**: proxy data and reconstructed inputs resemble noise and fail to reveal meaningful client information.
    - Studied **accuracy–communication–privacy trade-offs** by varying proxy size and update frequency, yielding practical guidelines for **bandwidth-constrained federated deployments**.

- **Event-Camera Object Detection** – *DOTIE* (ICRA 2023 & CVPR 2023 Workshop)
  - Developed an **event-camera** object detection pipeline using a lightweight **spiking layer** plus **density-based clustering** to isolate moving objects without frame reconstruction.
  - On the **MVSEC** outdoor driving dataset, more than doubled the mean IoU over prior event-based methods and achieved near-perfect foreground detection.
  - Achieved roughly **six orders of magnitude lower energy consumption** and significantly reduced latency compared to a **YOLO CNN** baseline on the same data.
  - Maintained performance across diverse scenes with minimal retuning, enabling deployment in **low-power** autonomous navigation and **neuromorphic** systems.

Work Experience

- **Research Intern, Integrated Systems Team** — Skillman, NJ
  Latent AI — *May 2023 – August 2023*
  - Prototyped an **unsupervised anomaly detection** pipeline for automated target recognition using state-of-the-art methods via Anomalib on MVTEC-AD and internal sensor datasets.
  - Built an **interactive labeling tool** combining classical computer vision and **SAM** to generate pixel-wise masks on noisy imagery, reducing manual annotation time.
  - Exported top models to ONNX and used the **Latent AI Efficient Inference Platform (LEIP)** to compile and optimize them for edge devices, to 4× **energy-efficiency gains** over unoptimized baselines.
  - Summarized experiments and findings in internal documentation and presentations to support ongoing edge inference pipeline development.

Selected Publications

- **Nagaraj, Manish**, Sakshi Choudhary, Utkarsh Saxena, Deepak Ravikumar, and Kaushik Roy. TRIM: Token-wise attention-derived saliency for data-efficient instruction tuning, 2025
- **Nagaraj, Manish**, Deepak Ravikumar, and Kaushik Roy. Coresets from trajectories: Selecting data via correlation of loss differences (CLD). *Transactions on Machine Learning Research (TMLR)*, 2025
- **Nagaraj, Manish**, Isha Garg, and Kaushik Roy. TOFU: Towards obfuscated federated updates by encoding weight updates into gradients from proxy data. *IEEE Access*, 2024
- **Manish Nagaraj**, Chamika Mihiranga Liyanagedera, and Kaushik Roy. DOTIE - detecting objects through temporal isolation of events using a spiking architecture. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pages 4858–4864, 2023
- Arjun Roy, **Manish Nagaraj**, Chamika Mihiranga Liyanagedera, and Kaushik Roy. Live demonstration: Real-time event-based speed detection using spiking neural networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4080–4081, 2023

Relevant Coursework

• Artificial Intelligence • Statistical Machine Learning • Random Processes and Probability • Linear Algebra • Computational Models and Algorithms(DSA) • Distributed Computer Systems

Skills

- **Programming & Scripting**: Python, Bash
- **Platforms**: Linux (Ubuntu, Red Hat), SLURM-based GPU clusters
- **Deep Learning & ML**: PyTorch, Hugging Face Transformers, NumPy, pandas, scikit-learn, OpenCV
- **Distributed & Large-Scale Training**: PyTorch DDP, DeepSpeed, TensorBoard
- **Tools & Packaging**: Docker, Git/GitHub, Conda